Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

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# Ass1: Create a Singly Linear Linked List with following operations
# Create Linked List
# Traverse and print the node values
# Insert node at a specific position
# Find Middle node and print its value
# Delete node
# Reverse list
# Calculate the sum of every two consecutive node values.

#creating a node with data and next.
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

#creating a class
class LinkedList:
    def __init__(self):
        self.head = None

    # 1. Create Linked List
    #function creation
    def create(self):
        n = int(input("Enter number of nodes: "))

        for i in range(n):
            data = int(input(f"Enter value for node {i + 1}: "))
            new_node = Node(data)

            if self.head is None:
                self.head = new_node
            else:
                temp = self.head
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                while temp.next:
                    temp = temp.next
                temp.next = new_node

        print("LL created successfully")


# 2. Traverse and Print

#function creation
def traverse(self):
    if self.head is None:
        print("LL is empty.")
        return

    temp = self.head
    while temp:
        print(temp.data, end=" -> ")
        temp = temp.next

    print("None")

# 3. Insert at Specific Position
#function creation
def insert(self, data, position):
    new_node = Node(data)

    if position < 1:
        print("Invalid position.")
        return

    if position == 1:
        new_node.next = self.head
        self.head = new_node
        return

    temp = self.head

    for i in range(position - 2):
        if temp is None:
            print("Invalid position.")
            return
        temp = temp.next
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    if temp is None:
        print("Invalid position.")
        return

    new_node.next = temp.next
    temp.next = new_node
    print("Node inserted successfully")

# 4. Find Middle Node
#function creation
def middle(self):
    if self.head is None:
        print("LL is empty.")
        return

    slow = self.head
    fast = self.head

    while fast and fast.next:
        slow = slow.next
        fast = fast.next.next

    print("Middle node value:", slow.data)

# 5. Delete Node at Specific Position
#function creation
def delete(self, position):
    if self.head is None:
        print("LL is empty.")
        return

    if position < 1:
        print("Invalid position.")
        return

    if position == 1:
        self.head = self.head.next
        print("Node deleted successfully")
        return

    temp = self.head
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                for i in range(position - 2):
            if temp is None:
                print("Invalid position.")
                return
            temp = temp.next

        if temp is None or temp.next is None:
            print("Invalid position.")
            return

        temp.next = temp.next.next
        print("Node deleted successfully")

    # 6. Reverse Linked List
    #function creation
    def reverse(self):
        previous = None
        current = self.head

        while current:
            next_node = current.next
            current.next = previous
            previous = current
            current = next_node

        self.head = previous
        print("LL reversed successfully")

    # 7. Sum of Every Two Consecutive Nodes
    #function creation

    def pair_sum(self):
        if self.head is None or self.head.next is None:
            print("At least two nodes are required.")
            return

        temp = self.head

        while temp and temp.next:
            total = temp.data + temp.next.data
            print(f"{temp.data} + {temp.next.data} = {total}")
            temp = temp.next
```

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# Main Program
ll = LinkedList() #ll: is object and LinkedList: is class

while True:
    print("1. Create LL")
    print("2. Traverse LL ")
    print("3. Insert Node")
    print("4. Find Middle Node")
    print("5. Delete Node")
    print("6. Reverse LL ")
    print("7. Sum of Two Consecutive Nodes")
    print("8. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        ll.create()

    elif choice == 2:
        ll.traverse()

    elif choice == 3:
        data = int(input("Enter value to insert: "))
        position = int(input("Enter position: "))
        ll.insert(data, position)

    elif choice == 4:
        ll.middle()

    elif choice == 5:
        position = int(input("Enter position to delete: "))
        ll.delete(position)

    elif choice == 6:
        ll.reverse()

    elif choice == 7:
        ll.pair_sum()

    elif choice == 8:
        print("Program terminated")
        break
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        else:
    print("Invalid choice please try again")
    print("LL created successfully")
```

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PS C:\Users\Admin\Desktop\DSA> & C:\Users\Admin\AppData\Local
ssignment/Assign_1/assignment1.py
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 1
Enter number of nodes: 5
Enter value for node 1: 2
Enter value for node 2: 45
Enter value for node 3: 91
Enter value for node 4: 73
Enter value for node 5: 61
LL created successfully
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 2
2 -> 45 -> 91 -> 73 -> 61 -> None
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
```

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PS C:\Users\Admin\Desktop\DSA> & C:\Users\Admin\AppData\Local
ssignment/Assign_1/assignment1.py
8. Exit
Enter your choice: 3
Enter value to insert: 98
Enter position: 2
Node inserted successfully
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 2
2 -> 98 -> 45 -> 91 -> 73 -> 61 -> None
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 4
Middle node value: 91
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 5
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PS C:\Users\Admin\Desktop\DSA> & C:\Users\Admin\AppData\Loca
ssignment/Assign_1/assignment1.py
8. Exit
Enter your choice: 5
Enter position to delete: 61
Invalid position.
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 2
2 -> 98 -> 45 -> 91 -> 73 -> 61 -> None
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 6
LL reversed successfully
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 2
61 -> 73 -> 91 -> 45 -> 98 -> 2 -> None
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```
Enter your choice: 2
61 -> 73 -> 91 -> 45 -> 98 -> 2 -> None
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 7
61 + 73 = 134
73 + 91 = 164
91 + 45 = 136
45 + 98 = 143
98 + 2 = 100
1. Create LL
2. Traverse LL
3. Insert Node
4. Find Middle Node
5. Delete Node
6. Reverse LL
7. Sum of Two Consecutive Nodes
8. Exit
Enter your choice: 8
Program terminated
PS C:\Users\Admin\Desktop\DSA>
```